

MA 301
Test 2
Spring - 2013

Problem	points	out of
1		10
2		8
3		16
4		21
5		21
Score		66

Name: SOLUTIONS

HR: _____

Signature: _____

Formula Sheet
* Let $z = re^{\theta i} = r(\cos \theta + i \sin \theta)$ - $z^n = re^{n\theta i} = r^n [\cos(n\theta) + i \sin(n\theta)]$ - $z^{1/n} = re^{\frac{\theta + 2\pi k}{n} i} = r^{1/n} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right]$ for $k = 0, 1, 2, \dots, n-1$ * If $\mathbf{F} = [M, N, P] \implies \text{curl}(\mathbf{F}) = [P_y - N_z, M_z - P_x, N_x - M_y]$

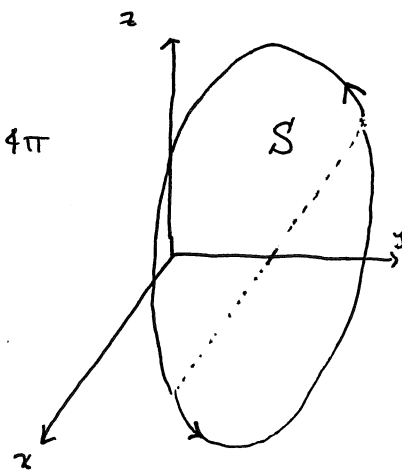
Problem 1: Let C be the boundary of the disc $x^2 + z^2 \leq 4$ in the $y = 1$ plane with positive orientation.

Compute $\oint_C \vec{F} \cdot d\vec{r}$ for the vector fields: (a) $\vec{F} = \left[\frac{M}{5z+2xz}, \frac{N}{1}, \frac{P}{x^2+y} \right]$ and (b) $\vec{F} = \left[\frac{M}{5zy}, \frac{N}{x^2}, \frac{P}{y} \right]$

$$(a) \quad \text{curl } \vec{F} = [1-0, 5+2x-2x, 0-0] = [1, 5, 0]$$

Since $\text{curl } \vec{F} \cdot \vec{n} = 5$ (constant) and $\text{Area}(S) = 4\pi$
then by Stokes' Thm

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S 5 dA = 5 \text{Area}(S) = 20\pi$$



$y=1$ plane

$$\vec{n} = [0, 1, 0]$$

$$\text{Area}(S) = 4\pi$$

$$(b) \quad \text{curl } \vec{F} = [1-0, 5y \quad -0, 2x-5z] \\ = [1, 5y, 2x-5z]$$

Since $\text{curl } \vec{F}$ is non-constant, let's parameterize S :

$$\begin{aligned} x &= r \cos \theta \\ y &= 1 \\ z &= r \sin \theta \end{aligned} \quad \xrightarrow[\substack{u=r \\ v=\theta}]{\substack{u=r \\ v=\theta}} \quad \vec{r}(u,v) = [u \cos v, 1, u \sin v] \quad \left. \begin{aligned} 0 \leq u \leq 2 \\ 0 \leq v \leq 2\pi \end{aligned} \right\} R$$

$$\begin{aligned} \Rightarrow \quad \vec{r}_u &= [\cos v, 0, \sin v] \\ \vec{r}_v &= [-u \sin v, 0, u \cos v] \end{aligned} \quad \Rightarrow \quad \vec{N} = [0, u \cos^2 v - u \sin^2 v, 0] \\ = [0, -u, 0] \quad \leftarrow \text{Note: } \vec{N} \text{ is in the wrong direction}$$

$$\text{curl } (\vec{F}(\vec{r})) = [1, 5, 2u \cos v - 5u \sin v]$$

$$\Rightarrow \text{curl } (\vec{F}(\vec{r})) \cdot (-\vec{N}) = 5u$$

so we should use $-\vec{N}$ for our normal.

$$\begin{aligned} \Rightarrow \oint_C \vec{F} \cdot d\vec{r} &= \iint_R 5u \, du \, dv = \int_0^{2\pi} \int_0^2 5u \, du \, dv \\ &= 2\pi \left(\frac{5}{2} u^2 \Big|_0^2 \right) \\ &= 2\pi \left(\frac{5}{2} (4) \right) = 20\pi \end{aligned}$$

Problem 2: Let $\mathbf{F} = \left[\frac{M}{xz-y}, \frac{N}{2yz-z}, \frac{P}{\frac{1}{2}x^2+y^2} \right]$ and C be the boundary of the quarter cone: $y^2 = x^2 + z^2$ with $x \geq 0, z \geq 0, 0 \leq y \leq 1$

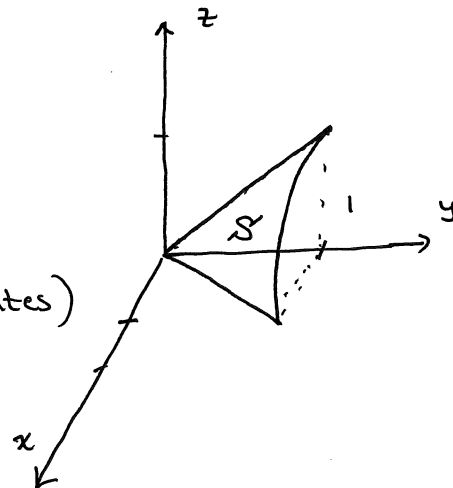
Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

$$\begin{aligned} \text{curl } \vec{F} &= [2y - (2y-1), x-x, 0+1] \\ &= [1, 0, 1] \quad (\text{constant}) \end{aligned}$$

Let's parameterize S (using cylindrical coordinates)

$$\begin{aligned} x &= y \cos \theta & u &= y \\ y &= y & \longrightarrow & \vec{r}(u, v) = [u \cos v, u, u \sin v] \\ z &= y \sin \theta & v &= \theta \end{aligned}$$

$$\left. \begin{aligned} 0 \leq u \leq 1 \\ 0 \leq v \leq \pi/2 \end{aligned} \right\} R$$



$$\begin{aligned} \vec{r}_u &= [\cos v, 1, \sin v] \\ \vec{r}_v &= [-u \sin v, 0, u \cos v] \end{aligned} \Rightarrow \vec{N} = [u \cos v, -u \cos^2 v - u \sin^2 v, u \sin v]$$

$$= [u \cos v, -u, u \sin v]$$

$$\text{curl } \vec{F}(\vec{r}) = [1, 0, 1]$$

$$\Rightarrow \text{curl } \vec{F}(\vec{r}) \cdot \vec{N} = u \cos v + u \sin v$$

$$\begin{aligned} \Rightarrow \oint_C \vec{F} \cdot d\vec{r} &= \int_0^{\pi/2} \int_0^1 u(\cos v + \sin v) du dv \\ &= \int_0^{\pi/2} (\cos v + \sin v) dv \cdot \int_0^1 u du \\ &= (\sin v - \cos v) \Big|_0^{\pi/2} \cdot \frac{u^2}{2} \Big|_0^1 \\ &= \left[\left(\sin\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi}{2}\right) \right) - \left(\sin(0) - \cos(0) \right) \right] \cdot \frac{1}{2} \\ &= [(1-0) - (0-1)] \cdot \frac{1}{2} \\ &= (1+1) \cdot \frac{1}{2} = 1 \end{aligned}$$

Problem 3: Answer the following conceptual questions:

(a) Suppose the series $\sum_{n=1}^{\infty} a_n$ converges and $a_n > 0$ for all n , determine the convergence of the following:
as $a_n \rightarrow 0^+$

(i) $\sum_{n=1}^{\infty} \ln(a_n)$ Since $\ln(a_n) \rightarrow -\infty \neq 0$ the series diverges

$a_n \rightarrow 0^+$

(ii) $\sum_{n=1}^{\infty} \ln(a_n + 1)$ Since $\ln(a_n + 1) \rightarrow 0$ as $a_n \rightarrow 0^+$ we cannot say anything about the convergence.

(iii) $\sum_{n=1}^{\infty} \cos(a_n)$ Since $\cos(a_n) \rightarrow 1$ as $a_n \rightarrow 0^+$ the series diverges.

(b) Suppose $\sum_{k=0}^{\infty} \left(\frac{4c}{5}\right)^k = \frac{5}{c^2}$. Find all values of c which satisfy this expression.

Geometric series that converges when $-1 < \frac{4c}{5} < 1 \Leftrightarrow -\frac{5}{4} < c < \frac{5}{4}$
and has the sum: $\frac{1}{1 - \frac{4c}{5}} = \frac{1}{\frac{5-4c}{5}} = \frac{5}{5-4c}$ from the G.S.T.

Equating the 2 values of the sum:

$$\frac{5}{5-4c} = \frac{5}{c^2} \Leftrightarrow 5-4c = c^2 \Leftrightarrow c^2 + 4c - 5 = 0$$

$$\Leftrightarrow (c-1)(c+5) = 0$$

$$\Leftrightarrow c = 1, -5 \leftarrow \text{Not in interval}$$

$$\boxed{c = 1}$$

(c) Suppose that the nth partial sum of the series $\sum_{k=1}^{\infty} a_k$ is $s_n = \frac{4n+6}{-1n+10}$.

(i) Find the first 3 partial sums

$$s_1 = \frac{4(1)+6}{-(1)+10} = \frac{10}{9}, \quad s_2 = \frac{4(2)+6}{-(2)+10} = \frac{14}{8}, \quad s_3 = \frac{4(3)+6}{-(3)+10} = \frac{18}{7}$$

(ii) Find a_1, a_2, a_3 (Note: $s_1 = a_1$, $s_2 = a_1 + a_2$, $s_3 = a_1 + a_2 + a_3$)

$$a_1 = s_1 = \frac{10}{9}, \quad a_2 = s_2 - a_1 = \frac{14}{8} - \frac{10}{9}, \quad a_3 = s_3 - \overbrace{(a_1 + a_2)}^{s_2}$$

$$= \frac{18}{7} - \frac{14}{8}$$

(iii) Does $\sum_{k=1}^{\infty} a_k$ converge? If so, what is its value?

Yes, this series converges since $\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \lim_{n \rightarrow \infty} s_n$

$$= \lim_{n \rightarrow \infty} \frac{4n+6}{-n+10} = -4$$

The sum is finite.

Problem 4: Determine the convergence of each series. If convergent, find the sum. Justify all work.

$$(a) \sum_{n=1}^{\infty} \frac{n^2}{n-n^2} \quad \lim_{n \rightarrow \infty} \frac{n^2}{n-n^2} = -1 \neq 0 \quad \sum a_n \text{ diverges}$$

$$(b) \sum_{n=1}^{\infty} \frac{4}{3^n} = 4 \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = 4 \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^{n+1} = \frac{4}{3} \left(\frac{1}{1-\frac{1}{3}}\right) = \frac{4}{3} \cdot \frac{3}{2} = 2$$

converges since $|r| < 1$

$$(c) \sum_{n=1}^{\infty} \frac{(-5)^{n-1}}{8^n} = -\frac{1}{5} \sum_{n=1}^{\infty} \left(-\frac{5}{8}\right)^n = -\frac{1}{5} \sum_{n=1}^{\infty} \left(-\frac{5}{8}\right)^{n+1} = -\frac{1}{5} \cdot -\frac{5}{8} \left(\frac{1}{1+\frac{5}{8}}\right)$$

converges since $|r| < 1$

$$= \frac{1}{8} \left(\frac{8}{13}\right) = \frac{1}{13}$$

$$(d) \sum_{n=1}^{\infty} \frac{\pi^{n+2}}{3^n} = \pi^2 \sum_{n=1}^{\infty} \left(\frac{\pi}{3}\right)^n \quad \frac{\pi}{3} > 1 \Rightarrow \text{diverges}$$

$$(e) \sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right) \quad \lim_{n \rightarrow \infty} \cos\left(\frac{1}{n^2}\right) = 1 \neq 0 \quad \text{diverges}$$

$$(f) \sum_{n=0}^{\infty} \frac{3^n}{10^{2n+1}} = \frac{1}{10} \sum_{n=0}^{\infty} \left(\frac{3}{100}\right)^n = \frac{1}{10} \left(\frac{1}{1-\frac{3}{100}}\right) = \frac{1}{10} \left(\frac{100}{97}\right) = \frac{10}{97}$$

converges since $|r| < 1$

$$(g) \sum_{n=4}^{\infty} \ln(n-1) - \ln(n-2)$$

$$\sum_{n=4}^N \ln(n-1) - \ln(n-2) = (\ln(3) - \ln(2)) + (\ln(4) - \ln(3)) + (\ln(5) - \ln(4)) + \dots + (\ln(N-2) - \ln(N-3)) + (\ln(N-1) - \ln(N-2))$$

$$= -\ln(2) + \ln(N-1)$$

$$\sum_{n=4}^{\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=4}^N a_n = \lim_{N \rightarrow \infty} (-\ln(2) + \ln(N-1)) = \infty$$

Diverges

Problem 5: Let $z_1 = 1 - i$, $z_2 = 3i$. Compute the following and leave answer in standard form.

(a) $z_1 - iz_2 = 1 - i - i(3i) = 1 - i + 3 = \boxed{4 - i}$

(b) $z_1 z_2 = (1 - i) 3i = 3i + 3 = \boxed{3 + 3i}$

(c) $|z_2| = \sqrt{9 + 0} = \boxed{3}$

(d) $\frac{z_2}{z_1 - z_2} = \frac{3i}{1 - i - 3i} = \frac{3i}{1 - 4i} \cdot \frac{1 + 4i}{1 + 4i} = \frac{3i - 12}{1 + 16} = \boxed{-\frac{12}{17} + \frac{3}{17}i}$

Compute the following and leave answer in polar form.

(e) Find the polar form of z_1 .

$$r = \sqrt{2} \quad \Rightarrow \quad z_1 = \sqrt{2} e^{-\frac{\pi}{4}i}$$

$$\theta = -\frac{\pi}{4}$$



(f) Find z_1^{12} .

$$z_1^{12} = (\sqrt{2})^{12} (e^{-\frac{\pi}{4}i})^{12} = 2^6 \cdot e^{-3\pi i}$$

(g) Find $z_1^{1/2}$. $z_1^{1/2} = (\sqrt{2})^{1/2} e^{\frac{-\frac{\pi}{4} + 2\pi k}{2}i}$ $k=0,1$

$$k=0: = 2^{1/4} e^{-\frac{\pi}{8}i}$$

$$k=1: = 2^{1/4} e^{\frac{7\pi}{8}i}$$

(h) Solve for x : $2x^3 - i = z_2$

$$2x^3 = 3i + i \Leftrightarrow x^3 = 2i \Leftrightarrow x = (2i)^{1/3} \quad \begin{cases} r = 2 \\ \theta = \frac{\pi}{2} \end{cases}$$

$$\text{So } x = 2^{1/3} e^{\frac{\frac{\pi}{2} + 2\pi k}{3}i} \quad k=0,1,2$$

$$k=0: = 2^{1/3} e^{\frac{\pi}{6}i}$$

$$k=1: = 2^{1/3} e^{\frac{5\pi}{6}i}$$

$$k=2: = 2^{1/3} e^{\frac{9\pi}{6}i}$$